

Fluid Flow in Pipe Systems

Final Lab Report
Unit Operations Lab 6
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1. Executive Summary

This experiment was performed to determine: the head loss for lengths of pipe of various materials, the loss coefficient of various pipe fittings and the flow coefficient for an orifice flowmeter. These parameters were calculated by measuring the differential pressure across each component at different flowrates, using a rotameter to verify flowrate. For the pipe fittings, we were able to determine the loss coefficients at each flowrate, but also that the loss coefficient decreases with increasing flowrate (or Reynold's Number) for all pipe fittings. This same relationship was also observed for the flow coefficient of the orifice and the friction factors (Darcy and Fanning), both values decreased with increasing flowrate. The significance of these results is that they show the pressure drop across all of these fluid system elements can be predicted with a high degree of accuracy, which allows pumps or compressors to be sized for a given fluid system given the components of the fluid system.

2. Introduction

The objective of this experiment is to examine and quantify the relationship between various parts of a fluid system (pipe lengths, pipe fittings and orifices) and the pressure drops across said parts. Almost every process of one type or another involves moving a fluid (liquid or gas) through a pipe and understanding how a fluid's pressure will drop along portions of a fluid system allows engineers to adequately size pumps/compressors to fit various system designs.

Our apparatus for this experiment consists of a water reservoir which leads to a centrifugal pump. Water exits the pump discharge and flows through a rotameter and into a single pipe with multiple types of pipe fitting in a series configuration (expansion,

contraction, smooth/ abrupt 90 degree turns, a smooth 180 degree turn and an orifice).

After these fittings there is a configuration 9 pipe lengths in a parallel configuration through which the water can be directed. There are 3 types of pipe: copper, stainless steel and plastic; arranged in groups of 3: large inner diameter, medium size inner diameter and small inner diameter with one of each type of metal in each group.

The industrial applications of this lab are almost limitless. Any application where a fluid is moved through a pipe is applicable: petrochemical production, natural gas delivery, potable water delivery, etc. The point is that if you want to move a fluid from point A to point B, you must know how much pressure the fluid will lose along the way.

3. Theory

The most basic theory on which this lab is based is derived from Newton's third law: for every action there is an equal and opposite reaction. (1) As fluid moves through a pipe, there is a force on the pipe due to the movement of the fluid. As a result, there is also a force exerted on the fluid by the pipe, or friction. This frictional force reduces the velocity of the fluid, which in turn reduces the pressure of the fluid. Bird uses this rationale as the basis of his derivation of equation 2.3-19 below:

$$v_z = \frac{(P_o - P_L)R^2}{4\mu L} \quad (2)$$

- Where v is the fluid velocity
- P is a summation of the pressure and gravitational terms
- R is the radius
- μ is the viscosity
- And L is the length of the pipe

For calculation of friction factors/flow coefficients/loss coefficients we use 2 methods different methods for calculation: Fanning Friction Factor (FFF) and Darcy

Friction Factor (DFF). The Fanning Friction Factor is calculated as follows:

$$f = \frac{1}{4} * \left(\frac{D}{L} \right) * \left(\frac{P_0 - P_L}{\frac{1}{2} * \rho <v>^2} \right) \quad [2] \quad (2)$$

- Where f is the Fanning Friction Factor
- P is previously defined
- <v> is the average fluid velocity
- ρ is the density of the fluid
- L is previously defined
- and D is the inner diameter of the pipe

However, the FFF can only be used for laminar flow. (3) To handle turbulent flow, we turn to the Darcy-Weisbach equation:

$$f = \left(\frac{D}{L} \right) * \left(\frac{P_0 - P_L}{\frac{1}{2} * \rho <v>^2} \right) \quad [3] \quad (3)$$

- Where f is the DFF
- All other variables have been previously defined

Pipe fitting friction losses depend on a factor (K) which is often experimentally determined and are usually published in data tables. (4) This friction loss, given in the form of fluid height, is calculated as follows:

$$h_L = K * \left(\frac{v^2}{2} \right) \quad [4] \quad (4)$$

- Where h_L is the head loss for the fitting

- K is the Friction Loss factor
- And v is previously fluid average velocity

K is defined as follows:

$$K = f\left(\frac{L}{D}\right) \quad [5] \quad (3)$$

- Where K is previously defined
- F is the friction factor
- And all other variables are previously defined

This can either be used to solve for K or rearranged to solve for f.

For a horizontally oriented orifice, the pressure drop across the orifice can be related to a change in fluid velocity using the Bernoulli's equation:

$$\Delta p = p_1 - p_2 = \frac{1}{2} * \rho (v_2^2 - v_1^2) \quad [6] \quad (5)$$

- Where all variables have been previously defined

The velocities can be replaced with areas by multiplying through by the volumetric flowrate of the fluid:

$$\Delta p = \frac{1}{2} \rho \dot{V}^2 * \frac{1}{A_2^2} * \left(1 - \frac{A_2^2}{A_1^2}\right) \quad [7] \quad (5)$$

- Where $\dot{V}$ is the volumetric flowrate of the fluid
- A is the cross-sectional area of the pipe
- And all other variables are previously defined

This equation is rearranged to solve for volumetric flowrate and a discharge coefficient is added to account for losses which convert kinetic energy into heat as follows:

$$\dot{V} = C_d \left(\frac{2\Delta p}{\rho}\right)^{1/2} * \frac{A_2}{\left(\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}\right)} \quad [8] \quad (5)$$

- Where C_d is the discharge coefficient
- And all other variables are previously defined

Because the flow of fluid immediately following the orifice is very complex, the effective area A_2 is unclear. This is accounted for by using a flow coefficient and the area of the orifice opening to substitute for terms in equation [8] as follows:

$$C_f * A_o = C_d * \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \quad [9] \quad (5)$$

- Where C_f is the flow coefficient
- A_o is the area of the orifice
- And all other variables are previously defined

Equation [8] can now be rewritten as:

$$\dot{V} = C_f A_o * \sqrt{2\Delta p / \rho} \quad [10] \quad (5)$$

- Where all variables have been previously defined

This equation can be rearranged to solve for the flow coefficient given the area of the orifice is known.

4. Results

In this lab there are 4 main objectives, which includes calibration of the rotameter, obtaining data for the orifice meter, and finding the head loss for the pipe friction and the pipe fittings. The other goal is to see how varying the pipe sizing and/or pipe material will change the flowrates due to friction.

Rotameter Calibration:

The average for the CF_{Rot} between the 3 trials was 0.8757. The given data for the lab states that this number should be 0.8455. This yields a 3.51% difference between the values, meaning the rotameter has not changed a significant amount.

Orifice Meter:

Orifice meters flow coefficient is calculated using the differential pressure across the plate, the volumetric flowrate, and the temperature of the water. (5)

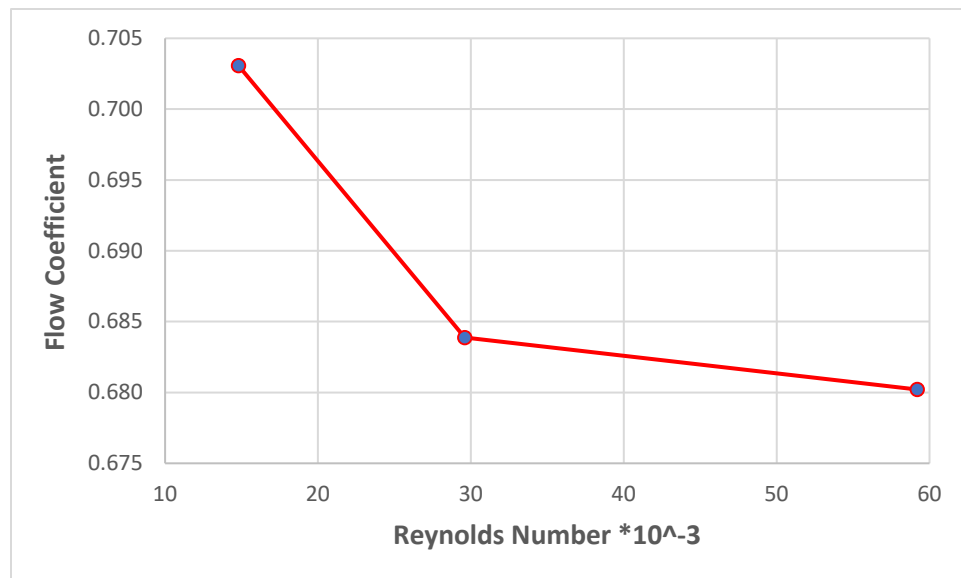


Figure 1: Orifice – Flow Coefficient vs Reynold's Number

Figure 4 shows the relationship of the orifice coefficient, or flow coefficient, compared to the Reynold's Number. The figure shows the three different flow rates that pushed through the orifice. The lowest flowrate yielded the largest flow coefficient.

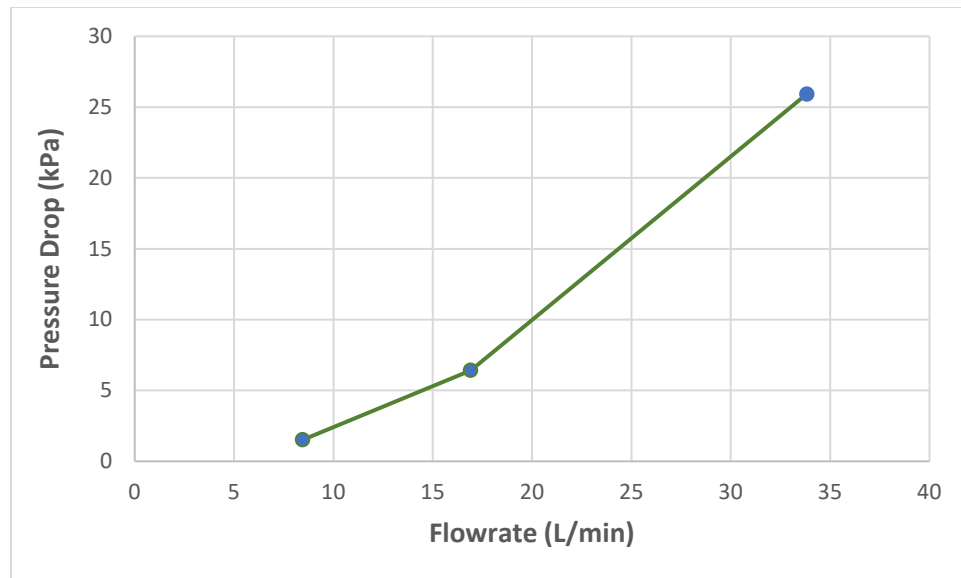


Figure 2: Orifice – Pressure Drop vs Flowrate

Figure 2 shows the direct relationship between pressure drop and flowrate. It is nearly a linear relationship and these values are directly proportional.

Pipe Friction Head Loss:

Using the three figures below, it is possible to compare the type of material, and size of material to one another. In all cases the blue line represents the small pipe diameter, the orange the medium, and the grey the large. The first point for the large piping all showed the same trend of being extremely high compared to that of the next flowrate.

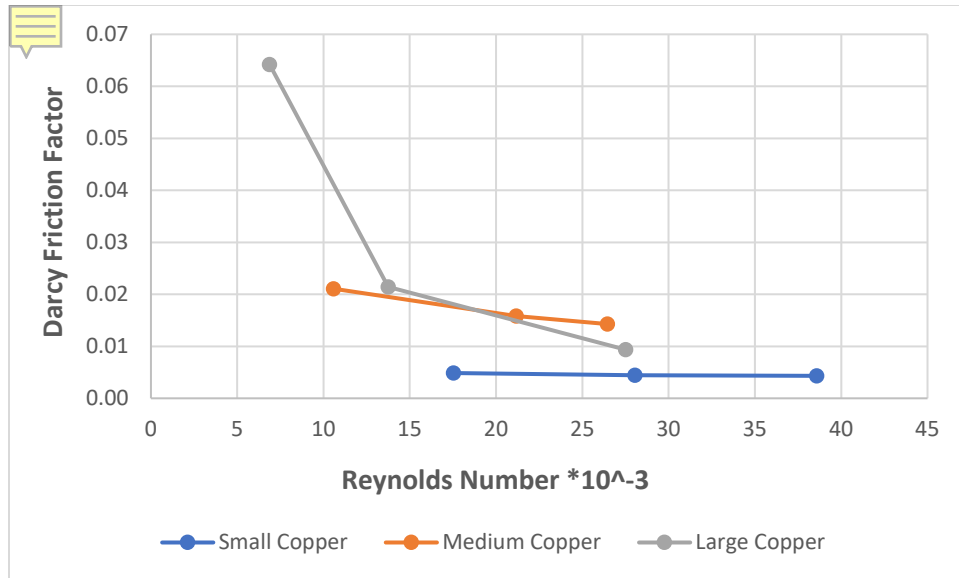


Figure 3: Copper – Friction Factor vs Reynold's Number (Experimental)

In figure 3 above the copper piping is investigated. The 3 points that are used are the three different flow rates that the trials were ran at, table 2 in Appendix I shows the flow rates for each material. The general trend that is shown is, as the friction factor decreased the Reynolds number increased. When looking at the small pipe it seems to have a no effect on the friction factor as the flowrates increased. Looking at the data in table 5 in Appendix I there is slight decrease in friction factor from 0.0049 to 0.0043. For the fastest flowrate, it is shown that the friction dips below that of the friction from the medium flowrate.

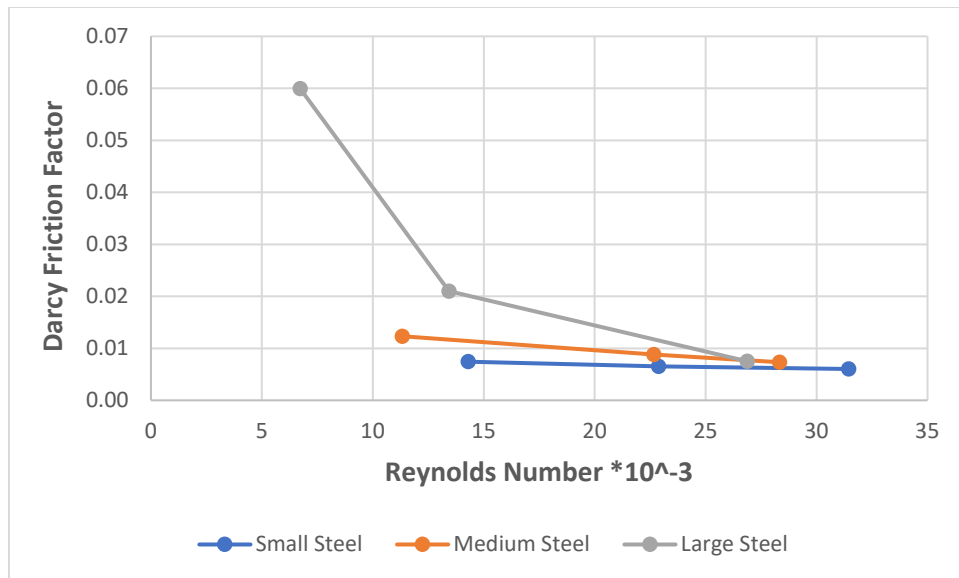


Figure 4: Steel – Friction Factor vs Reynold's Number

Steel follows the same trends as copper. However, the friction is much closer between the sizes, meaning there is not as much of a difference between the three pipe sizes as there is in the copper trial. Again, the large pipes friction decreases below the medium pipes' friction.

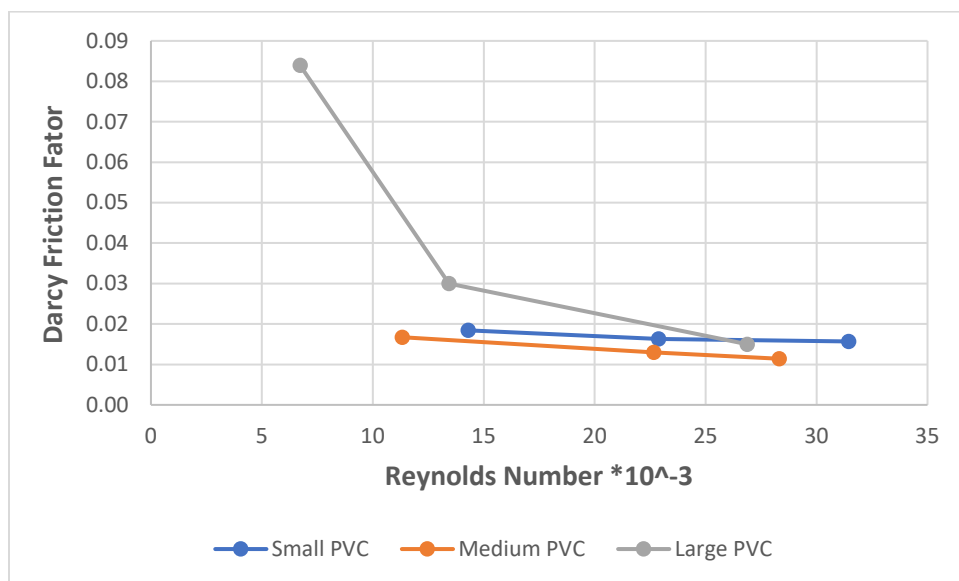


Figure 5: PVC – Friction Factor vs Reynold's Number

In figure 5, the PVC pipe was used. This yielded unorthodox values compared to that of the copper and Steel. The PVC had the medium flow have less friction than the small pipe did. As for the trends, all three sizes followed the same tendencies as the other two materials, with the low flowrate yielding the highest friction. The large piping shows the same trend dipping below the friction of another pipe; however, it has switched to the small pipe friction being the highest.

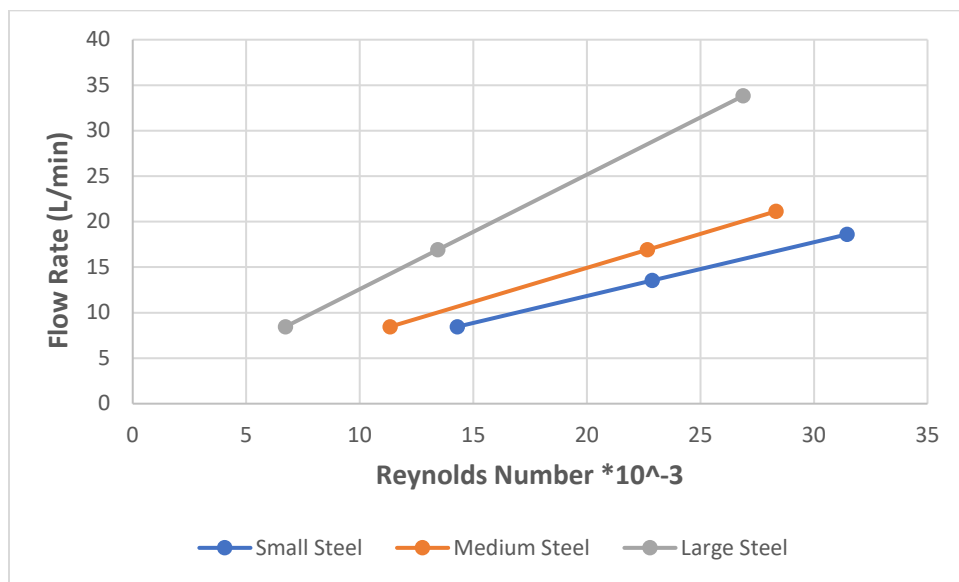


Figure 6: Steel – Flowrate vs Reynold's Number

Figure 6 shows the relationship of the Reynolds number and the flowrate in the steel piping. It was hypothesized that as flowrate increases so would the Reynolds number, this graph proves that theory.

Pipe Fittings Head Loss:

The focus of this section was to find the loss coefficient, K , for the different pipe fittings. Three flowrates were tested, refer to table 1 in appendix I. 5 different fittings were tested in the lab, figure 7 shows their trends.

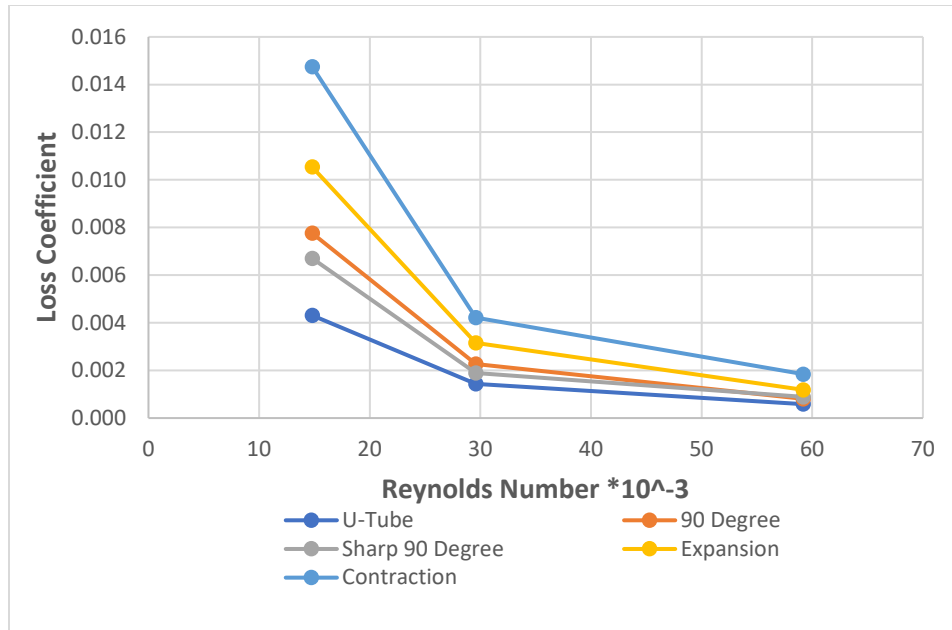


Figure 7: Fittings – Loss Coefficient vs Reynold's Number

In figure 7, as K decreases the Reynolds number tends to increase. All fittings followed the same trend and ended roughly around the same number for the loss coefficient in the fastest flowrate trial.

5. Discussion

The rotameter was within specifications at an average of 0.8757 being only 3.51 % difference from the given value of 0.8455.

The orifice meter was the only element that showed a significant pressure differential. This is the reason that venturi meters are preferred to orifice meters, since venturi will not have as bad of a differential pressure across the apparatus. From source (6) the literature value of the estimated orifice coefficient is 0.73 which is high for compared to what the experimental data provided, which was from .681 - .703. From source (7) which describes the pressure drop as linear and proportional to the flow rate of the liquid across

the orifice. The C_f , or flow coefficient, is not constant and does vary with flowrate, this is proven by looking at figure 1 in the results section.

From reference (8) the Reynolds number can be described in three ways, either as laminar, transient, or turbulent. Depending on the size Reynolds number changes what type of flow is shown:

- a. laminar - $Re < 2300$
- b. transient - $2300 < Re < 4000$
- c. turbulent - $Re > 4000$ (8)

The data shows that the smallest Re that is given is 6720.0 meaning all flows are turbulent for this lab. Meaning that there is no proof that this study would be able to cover the laminar flow. The known difference between the Fanning and the Darcy friction factor is that the Darcy is four times that of the Fanning. This is shown in the data, using table 6 for the small steel, shows the fanning factor being 0.0015 and the Darcy being 0.0060. The fanning friction factor is used to calculate laminar flow, where the Darcy factor is to solve for turbulent.

The fittings all followed the same trend, the largest disparity would be the contraction pipe for the slow flow rate. The contraction pipe also did not converge to the same point being about .005 higher than the rest. Leading the fact that the pipes all have roughly the same length as one another, with the higher the loss coefficient meaning the diameter is the varying factor. The longer the pipe the more chances it must accumulate the friction and loss pressure and flowrate.

In conclusion, the faster the fluid moved through the pipe, in any system, the less friction occurred. This follows the theory that was believed to be true.

6. References

- (1) Smith, Julios O. *Newton's Three Laws of Motion*
- (2) Bird, R., Stewart, W. and Lightfoot, E. (2007). *Transport Phenomena*. New York: John Wiley.
- (3) Darcy-Weisbach Formula Handout
- (4) Friction-Loss Fitting Handout
- (5) Unit Operations Lab: *Fluid Flow in Pipe Systems*. Dr. Zdunek. Fall 2019. Department of Chemical Engineering. University of Illinois at Chicago. Lab Manual.
- (6) Miller, Richard W (1996). *Flow Measurement Engineering Handbook*. New York: McGraw-Hill.
- (7) "Orifice Meter." *Scharf Automation*, 3 Oct. 2015.
- (8) "Reynolds Number." *Engineering ToolBox*.

7. Appendix I: Data Tabulation/Graphs

Table 1: Raw Data for Pipe Fittings

Fitting Type	Flow Rate (ft ³ /min)	Differential Pressure (psi)	Temperature (F)
Orifice	10	0.22	73.6
	20	0.93	74.3
	40	3.76	74.8
Sharp 90 Degree (P-6D)	10	0.08	73.5
	20	0.09	74.2
	40	0.17	74.8
Contraction (P-6B)	10	0.07	73.4
	20	0.08	74.2
	40	0.14	74.8
Expansion (P-6A)	10	0.05	73.3
	20	0.05	74.1
	40	0.09	74.7
90 Degree (P-5)	10	0.06	73.8
	20	0.07	74.4
	40	0.10	74.9
U-Tube (P-4)	10	0.06	73.9
	20	0.08	74.4
	40	0.13	74.9

Table 2: Raw Data for Pipe Materials

Fitting Material/Size	Flow Rate (ft ³ /min)	Differential Pressure (psi)	Temperature (F)
Steel Small	10	0.27	76.5
	16	0.61	76.5
	22	1.06	76.5
Steel Medium	10	0.14	75.8
	20	0.40	75.9
	25	0.52	75.9
Steel Large	10	0.05	75.6
	20	0.07	75.2
	40	0.10	75.1
Copper Small	10	0.49	76.5
	16	1.14	76.4
	22	2.11	76.4
Copper Medium	10	0.17	75.8
	20	0.51	75.8

	25	0.72	76.0
Copper Large	10	0.06	75.4
	20	0.08	75.3
	40	0.14	75.1
PVC Small	10	0.67	76.5
	16	1.52	76.4
	22	2.76	76.5
PVC Medium	10	0.19	75.7
	20	0.59	75.9
	25	0.81	76.1
PVC Large	10	0.07	75.3
	20	0.10	75.3
	40	0.20	75.0

Table 3: Results for Calculating Pipe's Reynolds Number

Fitting Material/Size	Flow Rate (L/min)	Differential Pressure (psi)	Temperature (C)	Density (kg/m ³)	Hydraulic Radius (m)	Pipe Area (m ²)	Velocity (m/s)	Pressure Drop (Pa)	Reynolds Number
Steel Small	18.60	0.27	24.72	997.15	0.0031	0.00012	2.52	7308.4	31453.5
	13.53	0.61	24.72	997.15	0.0031	0.00012	1.83	4205.8	22875.3
	8.46	1.06	24.72	997.15	0.0031	0.00012	1.15	1861.6	14297.1
Steel Medium	21.14	0.14	24.39	997.23	0.0040	0.00020	1.80	3585.3	28325.1
	16.91	0.40	24.39	997.23	0.0040	0.00020	1.44	2757.9	22660.0
	8.46	0.52	24.34	997.24	0.0040	0.00020	0.72	965.3	11330.2
Steel Large	33.82	0.05	24.28	997.26	0.0067	0.00056	1.01	689.5	26879.8
	16.91	0.07	24.22	997.27	0.0067	0.00056	0.51	482.6	13440.1
	8.46	0.10	24.22	997.27	0.0067	0.00056	0.25	344.7	6720.0
Copper Small	18.60	0.49	24.67	997.16	0.0026	0.00008	3.79	14547.9	38570.4
	13.53	1.14	24.67	997.16	0.0026	0.00008	2.76	7860.0	28051.2
	8.46	2.11	24.72	997.15	0.0026	0.00008	1.72	3378.4	17531.8
Copper Medium	21.14	0.17	24.45	997.22	0.0042	0.00022	1.57	4964.2	26449.7
	16.91	0.51	24.34	997.24	0.0042	0.00022	1.25	3516.3	21160.4
	8.46	0.72	24.34	997.24	0.0042	0.00022	0.63	1172.1	10580.2
Copper Large	33.82	0.06	23.95	997.34	0.0065	0.00053	1.06	965.3	27501.4
	16.91	0.08	24.06	997.31	0.0065	0.00053	0.53	551.6	13750.3
	8.46	0.14	24.11	997.30	0.0065	0.00053	0.26	413.7	6875.1
PVC Small	18.60	0.67	24.72	997.15	0.0031	0.00012	2.52	19029.5	31453.5
	13.53	1.52	24.67	997.16	0.0031	0.00012	1.83	10480.0	22875.6
	8.46	2.76	24.67	997.16	0.0031	0.00012	1.15	4619.5	14297.3
PVC Medium	21.14	0.19	24.50	997.20	0.0040	0.00020	1.80	5584.8	28324.3
	16.91	0.59	24.39	997.23	0.0040	0.00020	1.44	4067.9	22660.0
	8.46	0.81	24.28	997.26	0.0040	0.00020	0.72	1310.0	11330.3

PVC Large	33.82	0.07	23.89	997.35	0.0067	0.00056	1.01	1379.0	26882.4
	16.91	0.10	24.06	997.31	0.0067	0.00056	0.51	689.5	13440.6
	8.46	0.20	24.06	997.31	0.0067	0.00056	0.25	482.6	6720.3

Table 4: Results for Fitting's Reynolds Number

Fitting Material/Size	Flow Rate (L/min)	Differential Pressure (psi)	Temperature (C)	Density (kg/m ³)	Hydraulic Radius (m)	Pipe Area (m ²)	Velocity (m/s)	Pressure Drop (Pa)	Reynolds Number
Orifice	33.82	0.22	23.78	997.38	0.0030	0.00011	4.90	25924.3	59187.3
	16.91	0.93	23.50	997.45	0.0030	0.00011	2.45	6412.1	29595.6
	8.46	3.76	23.11	997.54	0.0030	0.00011	1.23	1516.8	14799.2
Sharp 90 Degree (P-6d)	33.82	0.08	23.78	997.38	0.0030	0.00011	4.90	1172.1	59187.3
	16.91	0.09	23.45	997.46	0.0030	0.00011	2.45	620.5	29596.0
	8.46	0.17	23.06	997.55	0.0030	0.00011	1.23	551.6	14799.4
Contraction (P-6B)	33.82	0.07	23.78	997.38	0.0030	0.00011	4.90	965.3	59187.3
	16.91	0.08	23.45	997.46	0.0030	0.00011	2.45	551.6	29596.0
	8.46	0.14	23.00	997.57	0.0030	0.00011	1.23	482.6	14799.6
Expansion (P-6a)	33.82	0.05	23.72	997.39	0.0030	0.00011	4.90	620.5	59188.1
	16.91	0.05	23.39	997.48	0.0030	0.00011	2.45	413.7	29596.4
	8.46	0.09	22.95	997.58	0.0030	0.00011	1.23	344.7	14799.8
90 Degree (P-5)	33.82	0.06	23.84	997.37	0.0030	0.00011	4.90	689.5	59186.5
	16.91	0.07	23.56	997.43	0.0030	0.00011	2.45	482.6	29595.2
	8.46	0.10	23.22	997.52	0.0030	0.00011	1.23	413.7	14798.8
U-Tube (P-4)	33.82	0.06	23.84	997.37	0.0030	0.00011	4.90	896.3	59186.5
	16.91	0.08	23.56	997.43	0.0030	0.00011	2.45	551.6	29595.2
	8.46	0.13	23.28	997.50	0.0030	0.00011	1.23	413.7	14798.6

Table 5: Calculating Pipe's Friction Factors and Head Loss

Material/Size	Flow Rate (L/min)	Fanning Friction Factor	Darcy-Weisbach Friction Factor	Head Loss (m)
Steel Small	18.60	0.0015	0.0060	0.1868
	13.53	0.0016	0.0066	0.1075
	8.46	0.0019	0.0074	0.0476
Steel Medium	21.14	0.0018	0.0073	0.0916
	16.91	0.0022	0.0088	0.0705
	8.46	0.0031	0.0123	0.0247
Steel Large	33.82	0.0019	0.0075	0.0176
	16.91	0.0052	0.0210	0.0123
	8.46	0.0150	0.0600	0.0088
Copper Small	18.60	0.0011	0.0043	0.3718
	13.53	0.0011	0.0044	0.2009
	8.46	0.0012	0.0049	0.0863
Copper Medium	21.14	0.0036	0.0143	0.1269
	16.91	0.0040	0.0158	0.0899
	8.46	0.0053	0.0211	0.0300
Copper Large	33.82	0.0023	0.0094	0.0247
	16.91	0.0054	0.0214	0.0141
	8.46	0.0161	0.0642	0.0106
PVC Small	18.60	0.0039	0.0157	0.4863
	13.53	0.0041	0.0163	0.2678
	8.46	0.0046	0.0184	0.1181
PVC Medium	21.14	0.0029	0.0114	0.1427
	16.91	0.0032	0.0130	0.1040
	8.46	0.0042	0.0167	0.0335
PVC Large	33.82	0.0037	0.0150	0.0352
	16.91	0.0075	0.0300	0.0176
	8.46	0.0210	0.0840	0.0123

Table 6: Calculating Fitting's Friction Factors and Loss Coefficients

Fitting Material/Size	Flow Rate (L/min)	Fanning Friction Factor	Darcy-Weisbach Friction Factor	Loss Coefficient	Head Loss (m)
Sharp 90 Degree (P-6d)	33.82	0.000062	0.00025	0.0009	0.01071
	16.91	0.000130	0.00052	0.0019	0.00567
	8.46	0.000464	0.00185	0.0067	0.00504
Contraction (P-6B)	33.82	0.000051	0.00020	0.0018	0.02218
	16.91	0.000116	0.00046	0.0042	0.01267
	8.46	0.000406	0.00162	0.0148	0.01108
Expansion (P-6a)	33.82	0.000033	0.00013	0.0012	0.01426
	16.91	0.000087	0.00035	0.0032	0.00950
	8.46	0.000290	0.00116	0.0105	0.00792
90 Degree (P-5)	33.82	0.000036	0.00014	0.0008	0.00972
	16.91	0.000101	0.00041	0.0023	0.00680
	8.46	0.000348	0.00139	0.0078	0.00583
U-Tube (P-4)	33.82	0.000047	0.00019	0.0006	0.00702
	16.91	0.000116	0.00046	0.0014	0.00432
	8.46	0.000348	0.00139	0.0043	0.00324

Table 7: Calculating Orifice's Friction Factors and Flow Coefficients

Fitting	Flow Rate (L/min)	Fanning Friction Factor	Darcy-Weisbach Friction Factor	Flow Coefficient
Orifice	33.82	0.001362	0.00545	0.6802
	16.91	0.001347	0.00539	0.6839
	8.46	0.001275	0.00510	0.7031

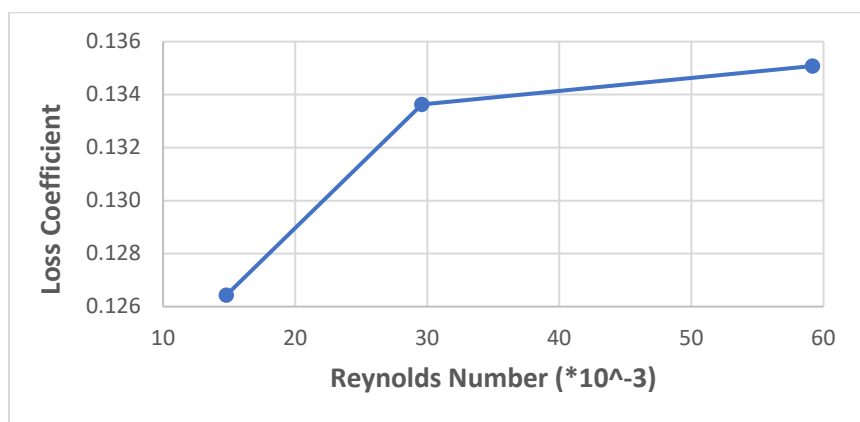


Figure 8: Orifice – Loss Coefficient vs Reynold's Number

8. Appendix II: Error Analysis

There were several factors within the experiment that could have contributed to error in the lab. One of the first sources of error was the heavy leaking of water throughout the pipe system. This error was due to having older pipes and valves being used in the experiment, in which must be changed out if they are leaking extensively throughout the system. That loss of water could contribute to some error, as there were many points of leaking that could not be accounted for that can affect the results. Another source of error in this lab came from the pressure and temperature readings. At each trial, each reader would never remain steady and fluctuate at about (± 0.2). Having this amount of error for our readings will impact some of the results slightly. The last source of error in the lab was the flow rate of water. Before beginning the experiment, the rotameter had to be calibrated to ensure the flow conversion of standard cubic feet per minute to liter per minutes equaled 0.8455. Any off value for the conversion factor will mean that there may be some inconsistency in the amount of actual flow in the system versus what's being read from the flowmeter. After careful trials in attempting to find the conversion of the rotameter, it was found to be about 0.8757, which results in about a ~6% error versus the theoretical conversion. The table below displays the trials that were done to find the rotameter conversion and finding the percent error.

Table 8: Error Analysis in Rotameter Conversion

Flow Rate (ft ³ /min)	Flow Rate (L/min)	Trials	Volume (L)	Time (sec)	Experimental Conversion	Theoretical Conversion	Average Percent Error (%)
10	8.455	1	0.965	7.676	0.8920	0.8455	6.06%
		2	0.965	7.434	0.9211		
		3	0.965	8.414	0.8139		

9. Appendix III: Sample Calculations

Rotameter Calibration

CF_{Rot} = Correction Factor for Rotameter, t = time, V = Volume Collected,
 Q_r = Theoretical Volume Rate Read from Rotameter

$$CF_{Rot} = \frac{\frac{V}{t}}{Q_r}$$

$$CF_{Rot} = \frac{\frac{0.965L}{7.67 \frac{sec}{min}} * \frac{60 \frac{sec}{min}}{min}}{\frac{8.455L}{min}}$$

$CF_{Rot} = 0.8920$

Reynold's Number

Data from Using Medium Sized Steel Pipe at 25 ft³/min

v = Velocity, ρ = Density of Water, D = Pipe Diameter, m = Viscosity of Water

$$Re = \frac{D * v * \rho}{m}$$

$$Re = \frac{0.01588m * 1.798 \frac{m}{s} * 997.23 \frac{kg}{m^3}}{0.001 Pa * s}$$

$Re = 28325.05$

Pipe Friction Factor and Head Loss

Data from Using Medium Sized Steel Pipe at 25 ft³/min

f = Darcy Friction Factor, ΔP = Pressure Drop, L = Length of Pipe,

$$f = \frac{\Delta P * D}{2 * \rho * L * v^2}$$

$$f = \frac{3585.3 \text{ Pa} * 0.01580 \text{ m}}{2 * 997.22 \frac{\text{kg}}{\text{m}^3} * 1.2 \text{ m} * (1.798 \frac{\text{m}}{\text{s}})^2}$$

$$f = 0.00732$$

$$h_L = f \frac{L V^2}{D 2}$$

$$h_L = 0.00732 * \frac{1.2 \text{ m}}{0.01580 \text{ m}} * \frac{(1.798 \frac{\text{m}}{\text{s}})^2}{2 * 9.8 \frac{\text{m}}{\text{s}^2}}$$

$$h_L = 0.0916 \text{ m}$$

Orifice Flow Coefficient

Data from Using Orifice Meter at 40 ft³/min

C_f = Flow Coefficient, Q = Volumetric Flow Rate, A_o = Area of Orifice

$$C_f = \frac{Q}{A_o \sqrt{\frac{2 \Delta p}{\rho}}}$$

$$C_f = \frac{0.00564 \frac{\text{m}^3}{\text{s}}}{0.000115 \text{ m}^2 * \sqrt{\frac{2 * 25924.3 \text{ Pa}}{997.38 \text{ kg/m}^3}}}$$

$$C_f = 0.6802$$

Pipe Fitting: Loss Coefficient

K = Loss Coefficient, f_f = friction factor, L = Length of Pipe,
 D = Diameter of Pipe, v = Velocity

$$K = f_f * \frac{L}{D}$$

$$K = 0.0000362 * \frac{0.27 \text{ m}}{0.0120 \text{ m}}$$

$$K = 0.000803$$

10. Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8 Hours	All
Pre-Lab Editing	1 Hours	All
Experimental Objective (Prelab)	1 Hour	Caleb
Apparatus (prelab)	2 Hours	Mike
Materials & Supplies (prelab)	1 Hour	Mike
Procedure (prelab)	2 Hours	Emir
Project Safety Evaluation (prelab)	1 Hours	Caleb
Final Lab Editing	3 Hour	All
Executive Summary (final lab)	2 Hour	Caleb
Introduction (Final lab)	2 Hours	Caleb
Theory (Final Lab)	4 Hours	Caleb
Results (final lab)	6 Hours	Mike
Discussion (final lab)	3 Hours	Mike
References (final and/or prelab)	1 Hour	All
Data Tabulation/Graphs (final)	8 Hours	Emir
Error Analysis (final lab)	1 Hour	Emir
Sample Calculations (final Lab)	4 Hours	Emir
Power Point Presentation	2 Hours	All
Total	52 Hours	All